[SMART CHILD HUB LOGO PLACEHOLDER]

DESIGN SYSTEM

User Interface and User Experience Guide

Smart Child Daily Activity Tracking System

An Offline Family Dashboard for Monitoring Children's Daily Habits

Phase 2 – UI/UX Design System (Material 3 · Glassmorphism · Soft UI)

Prepared for the development team

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CHAPTER ONE

INTRODUCTION

1.1 Purpose of the Design System

The Smart Child Hub design system is the single source of truth for the user interface and user experience of the Smart Child Daily Activity Tracking System. It defines the visual language, design tokens, colour palettes, typography, spacing, motion, components, icons, illustrations and screen layouts that will be used to build the application in Flutter during Phase 3. Every pixel of every screen, every component and every animation shown in this document is specified with exact values so that implementation is deterministic and does not require creative interpretation.

The system deliberately blends three contemporary UI styles: a Material 3 foundation for consistency and accessibility, a glassmorphism treatment (translucent frosted panels over soft gradients) for a modern, layered feel, and soft UI (large rounded shapes, gentle shadows and friendly colours) so that the interface feels warm and approachable for young children. The result is a playful child mode and a professional parent mode that share one underlying token system.

1.2 Scope

This design system covers the complete visual and interaction design of the application. It includes two colour themes (light and dark), a 19-style type scale built from three fonts (Baloo 2, Quicksand and Nunito), a 4–64 px spacing scale, four elevation levels, six motion curves and seven reference durations, eleven component boards, twenty-two full-screen mockups, fifty-two custom SVG icons and five illustrations. The scope is strictly visual and interactional; no Flutter, PHP, MySQL or API implementation code is included in this phase.

1.3 How to Use This Guide

- **Designers** use the mockups and component boards to validate new screens against the visual language.
- **Developers** read the design tokens and widget guide to map every token to a Flutter theme before writing widgets.
- **Reviewers** check screens against the accessibility rules, contrast minimums and touch-target minimums defined in Chapter 3 and Chapter 9.

1.4 Deliverables Overview

1: *Phase 2 deliverables at a glance*

Deliverable	Location	Count / Format
Design tokens	scripts/tokens.py	1 source file
UI Style Guide	style-guide/UI_Style_Guide.pdf	15 pages
Color Palette	style-guide/Color_Palette.pdf	16 pages
Typography Guide	style-guide/Typography_Guide.pdf	11 pages
Component Library	style-guide/Component_Library.pdf	24 pages
Animation Guide	style-guide/Animation_Guide.pdf	13 pages
Responsive Guide	style-guide/Responsive_Guide.pdf	11 pages
Widget Guide	style-guide/Widget_Guide.pdf	13 pages
Component boards	components/	11 PNG boards
Screen mockups	mockups/	22 PNG screens
Icons	assets/icons/	52 SVG files
Illustrations	assets/illustrations/	5 SVG files
Screen specifications	design-system/specs/	docx + markdown
Navigation flow	design-system/specs/	docx + markdown

1.5 Design Personas and Modes

The interface has two distinct visual voices backed by one token system. Child mode is bright, big and joyful: Baloo 2 display type, the Sunny mascot, candy accents, large cards and confetti. Parent mode is calm and professional: Nunito type, tighter spacing, data-dense charts and tabular numerals. The switch is automatic based on the active profile, never a setting the child can change.

2: *Child mode versus parent mode*

Characteristic	Child Mode (My Day)	Parent Mode (Family Manager)
Display type	Baloo 2 (700–800)	Nunito (600–800)
Body type	Quicksand (500–700)	Nunito (400–600)
Numerals	Quicksand tabular	Nunito tabular
Layout density	Spacious (16–32 px)	Compact (8–16 px)
Navigation	Bottom bar	Navigation rail + app bar
Accents	Candy, sky, grape	Primary blue, success green
Motifs	Sunny, coins, stars	Charts, tables, approvals

CHAPTER TWO

DESIGN PRINCIPLES AND VISUAL LANGUAGE

2.1 The Four Design Principles

1. Playful – big rounded shapes, the friendly Sunny mascot and warm colours make routines feel like games, never chores. Everything in child mode is scaled, spaced and coloured for delight first.
2. Clear – one primary action per screen, big type and generous spacing mean a four-year-old and a parent both understand the screen instantly. No hidden gestures, no ambiguous icons.
3. Safe – parent-only zones sit behind a PIN. The app runs completely offline with no accounts, no ads and no internet dependency, keeping the family's data private inside the home.
4. Rewarding – coins, badges and confetti celebrate every win and turn effort into visible, motivating progress. Positive reinforcement is the core emotional loop.

2.2 Visual Language

- **Frosted glass (glassmorphism)** panels with soft translucent fills over a subtle gradient backdrop create depth without heavy borders.
- **Soft UI** rounded rectangles (8–32 px radii), low-contrast soft shadows and muted surface tints keep the interface gentle on the eye.
- **Gradients** reserved for hero areas, the splash, celebrations and the Sunny mascot; content surfaces stay clean and flat.
- **Illustration** the Sunny mascot and five supporting illustrations carry the personality; they never carry critical information.

2.3 The Mascot: Sunny

Sunny is a smiling sun used across the brand, the splash screen, empty states, celebrations and the child dashboard. Sunny is drawn in flat vector style with the brand colours, an orange glow, rosy cheeks and a friendly closed-mouth smile. The mascot animates with a gentle bobbing motion on idle screens and spins into the splash screen. Sunny is always a supporter, never a judge: on error states Sunny appears helpful and calm.

2.4 Voice and Tone

- **Child voice** short positive sentences, encouraging words (“Great job!”, “You did it!”), never shame or negative framing for missed tasks.
- **Parent voice** clear, factual and reassuring (“3 approvals pending”, “Backup completed”).
- **Error tone** constructive: explain what happened and the one action that fixes it, never blame the child.

CHAPTER THREE

COLOUR SYSTEM AND DESIGN TOKENS

3.1 Token Philosophy

All colours are stored as semantic tokens, not raw hex values. Screens and components reference token names such as primary, surface and success so that theming (light, dark, brand tweaks) is a single point of change. The complete token set is defined in scripts/tokens.py and rendered as reference boards in components/10_colors.png.

3.2 Brand Palette

3: Brand colour tokens (light theme)

Token	Light Value	Purpose
primary	#4F6BF5	Primary actions, selected states
onPrimary	#FFFFFF	Content on primary
primaryContainer	#DDE5FF	Soft primary backgrounds
onPrimaryContainer	#0E1E5C	Content on primary container
secondary	#FF8A3D	Secondary actions, highlights
secondaryContainer	#FFE0C7	Soft secondary backgrounds
tertiary	#1FBFA0	Verified/success accents, brand pop
background	#F7F8FE	Screen background (light)
surface	#FFFFFF	Cards and surfaces
onSurface	#191A24	Primary text
surfaceVariant	#E9EAF6	Dividers, subtle fills
onSurfaceVariant	#43434F	Secondary text
outline	#767685	Borders, icons at rest
outlineVariant	#D5D6E4	Light borders

3.3 Accent Colours

4: Accent tokens for avatars, categories and delight

Token	Value	Use
sunshine	#FFC93C	Coins, sparkles, celebration
candy	#FF7BA9	Child avatars, playful accents
sky	#54C3F6	Child avatars, weather-friendly tints
grape	#A98BFF	Badges, child avatars
mint	#3DDB9E	Success flourishes, fresh tints
peach	#FFB37A	Warm tints, secondary highlights

3.4 Semantic and Status Colours

Status is never communicated by colour alone; each task status pairs its colour with an icon and a text label. This keeps the interface accessible for colour-blind users.

5: The six task status colours

Status	Value	Icon	Meaning
completed	#2FAE66	check	Task done and verified
pending	#FF9F0A	clock	Waiting for parent approval
rejected	#E5484D	close	Submission rejected by parent
verified	#1FBFA0	shield	Self-checked and accepted
missed	#8E8E99	flag	Due date passed without completion
locked	#B9BAC6	lock	Task not yet available

3.5 Badge Tiers

6: Achievement badge tier colours

Tier	Foreground	Background	Feeling
bronze	#C98A5B	#F5E0D1	Getting started
silver	#9AA4B5	#E9EDF3	Consistent effort
gold	#E8A500	#FCEFC8	Outstanding streak
diamond	#3FB9E0	#D8F3FB	Elite achievement

3.6 Dark Theme

The dark theme keeps the exact same token structure and hue families but shifts luminance. Surfaces become deep indigo-greys, primary becomes a light periwinkle (B7C4FF) so it stays legible on dark surfaces, and error shifts to a lighter red (FFB4AB). The full dark palette is rendered in the Color Palette guide and on the colour board.

3.7 Contrast and Accessibility

- **Body text on surfaces** must reach WCAG AA (4.5:1); the standard onSurface (#191A24) on surface (#FFFFFF) reaches 16.2:1.
- **Large text (18+ px or 14 px bold)** must reach 3:1; all display and headline styles exceed this.
- **Status chips** always pair colour with icon and label, never rely on colour alone.
- **Focus states** a visible 2 px primary ring with 8 px offset on all interactive elements.

CHAPTER FOUR

TYPOGRAPHY

4.1 The Three-Family System

Three complementary open-source font families create clear hierarchy and instantly distinguish the two modes. Baloo 2 is the rounded, friendly display face for headlines and numbers in child mode. Quicksand is the soft geometric face for child-mode body text and interface numerals. Nunito is the neutral, highly legible face for all parent-mode text and data.

7: The three font families and their roles

Family	Role	Weights used
Baloo 2	Brand display, child headlines, hero numbers	400, 500, 600, 700, 800
Quicksand	Child body text, coins and numerals	400, 500, 600, 700
Nunito	Parent UI, body text, charts, labels	400, 500, 600, 700, 800

4.2 Type Scale

The scale defines nineteen styles. Every style fixes the family, size, line height and weight. Implementations must use these exact values, not eyeballed approximations.

8: The complete 19-style type scale

Style	Family	Size	Line	Weight	Use
Display Large	Baloo 2	57	60	800	Hero numbers, splash
Display Medium	Baloo 2	45	48	700	Screen hero titles (child)
Display Small	Baloo 2	36	40	700	Welcome, celebration
Headline Large	Baloo 2	32	36	700	Dashboard main heading
Headline Medium	Baloo 2	28	32	700	Section titles, headers
Headline Small	Baloo 2	24	28	700	Card titles, dialogs
Title Large	Nunito	22	26	700	Parent screen titles
Title Medium	Nunito	16	24	600	Card titles (parent)
Title Small	Nunito	14	20	600	List items, nav

					items
Body Large	Nunito	16	24	400	Primary body text
Body Medium	Nunito	14	20	400	Secondary text
Body Small	Nunito	12	16	400	Captions, meta
Label Large	Nunito	14	20	600	Buttons, tabs
Label Medium	Nunito	12	16	600	Chips, badges, labels
Label Small	Nunito	11	16	600	Overlines, timestamps
Stat Large	Nunito	48	52	800	Coin totals (tabular)
Stat Medium	Nunito	28	32	800	Dashboard statistics
Stat Small	Nunito	18	24	700	Coins on cards
Chart Label	Nunito	11	14	600	Axis labels, legends

4.3 Numerals and Tabular Figures

All numeric totals, statistics and coin values use tabular figures (fixed-width numerals) so that numbers align in tables and count-up animations do not jitter. Quicksand is preferred for child-mode numerals; Nunito for parent-mode data.

4.4 Typography Hygiene

- Never exceed two display words per line in child mode.
- Line height minimums from the scale are binding; do not compress for space.
- Text on glass panels uses the onSurface tokens, not pure white or black.
- Abbreviations and acronyms are spelled out in child mode (“coins” not “coins.”).

CHAPTER FIVE

LAYOUT, GRID AND SPACING

5.1 Breakpoints

9: *Responsive breakpoint tiers*

Tier	Width	Target
compact	Below 600 dp	Phone portrait – usable but not primary
medium	600–840 dp	Tablet portrait – the primary target
expanded	Above 840 dp	Tablet landscape and large screens

5.2 The Grid

- **Medium (600–840 dp)** a 6-column grid with 16 px gutters and 24 px screen padding.
- **Expanded (above 840 dp)** a 12-column grid, same 16 px gutters and 24 px padding, with a content max width of 880 dp.
- **Compact** a 4-column grid with 16 px gutters and 16 px padding.

5.3 Spacing Scale

10: *The 4–64 px spacing scale*

Token	Value	Use
4	4 px	Tight gaps between icon and label in small chips
8	8 px	Gaps between elements inside cards, icon padding
12	12 px	Default internal card padding, small gaps
16	16 px	Default screen padding, gaps between cards
20	20 px	Gaps between groups of related cards
24	24 px	Section spacing, dialog padding
32	32 px	Between major screen sections
40	40 px	Before/after a screen header or hero
48	48 px	Page gutters on large screens
64	64 px	Hero spacing, distinct screen zones

5.4 Radii and Elevation

11: *Corner radius tokens*

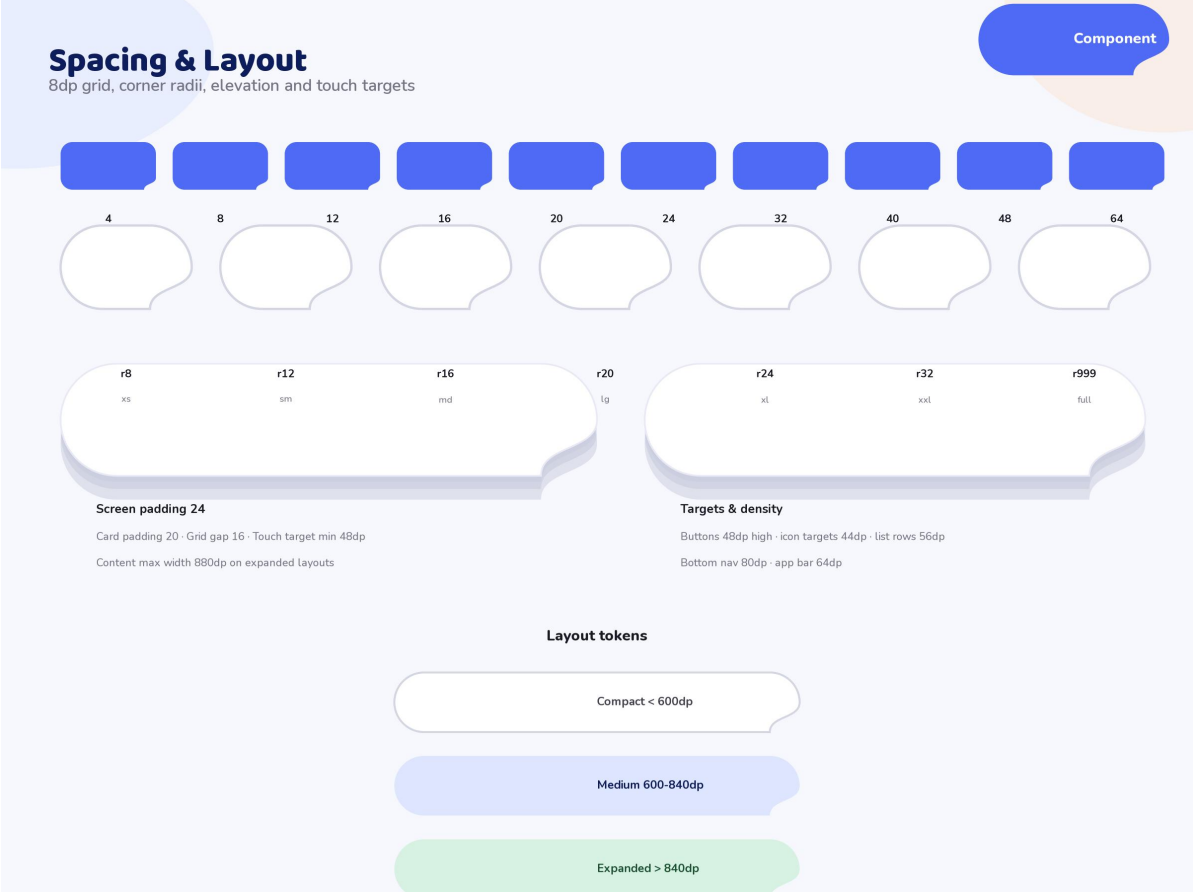
Token	Value	Use
xs / sm / md	8 / 12 / 16 px	Chips, small elements, inputs
lg / xl	20 / 24 px	Cards, dialogs, buttons
xxl / full	32 px / 999	Hero cards, FAB, avatar circles

12: *Elevation levels*

Level	Shadow	Blur	Alpha	Use
0	0	0	0.00	Resting flat elements
1	2	6	0.10	Cards at rest
2	4	12	0.14	Floating cards, FAB
3	8	24	0.18	Dialogs, bottom sheets
4	12	32	0.22	Overlays, snackbars

5.5 Touch Targets

Every interactive element has a minimum touch target of 48 × 48 dp. Visual size may be smaller, but the hit area never is. The bottom navigation bar is 80 dp tall and the app bar 64 dp tall.

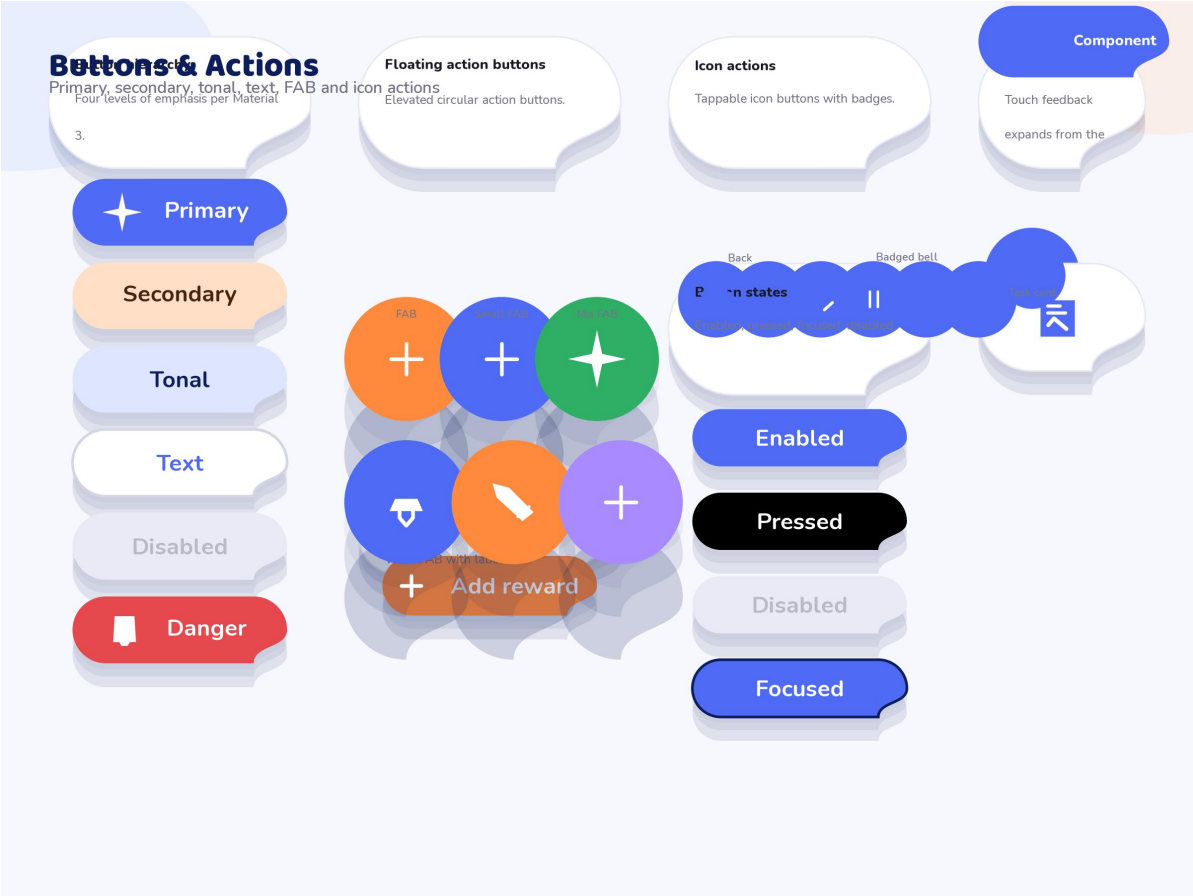


1: *Spacing, radius, elevation and grid reference board*

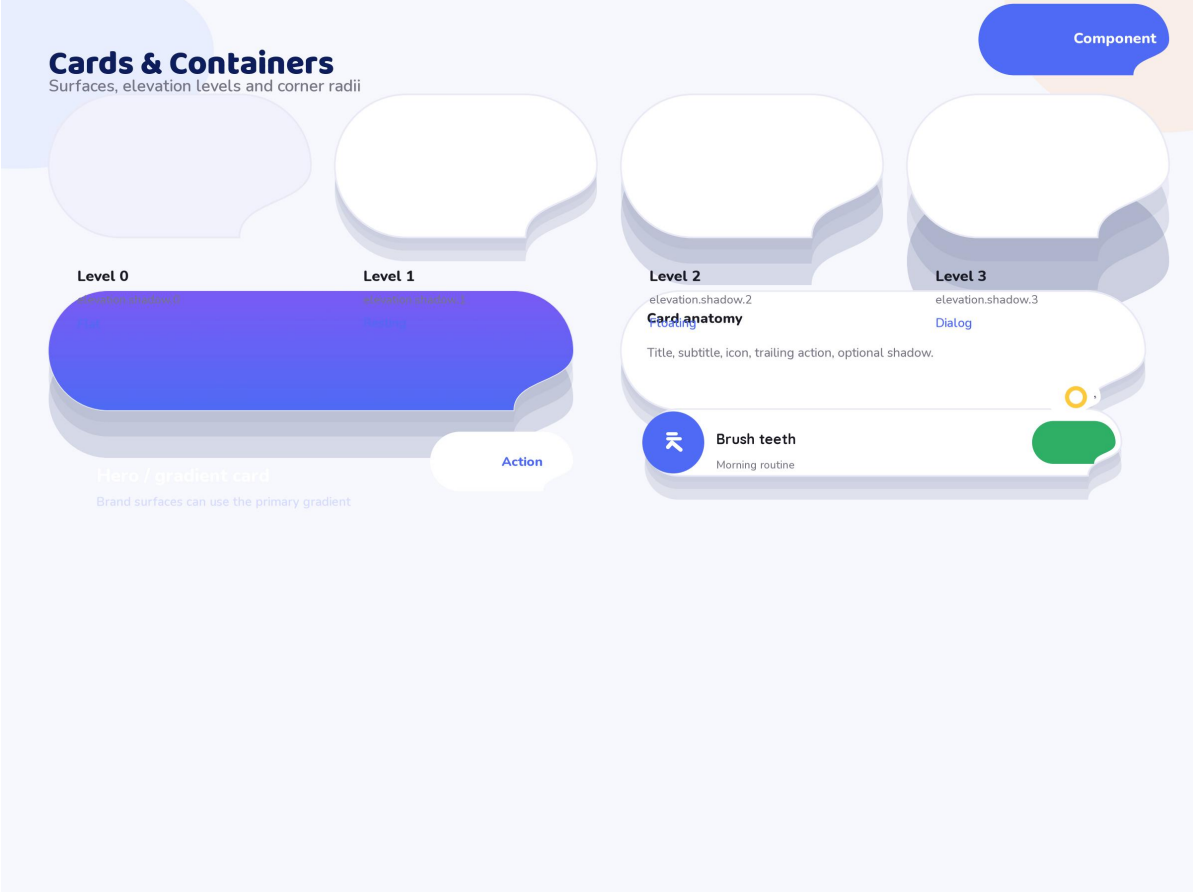
CHAPTER SIX

COMPONENT LIBRARY

The component library contains eleven reference boards. Each board is the single visual spec for its component family; the full sizing, state and usage rules live in the Component Library PDF. The boards below are the authoritative visual reference.



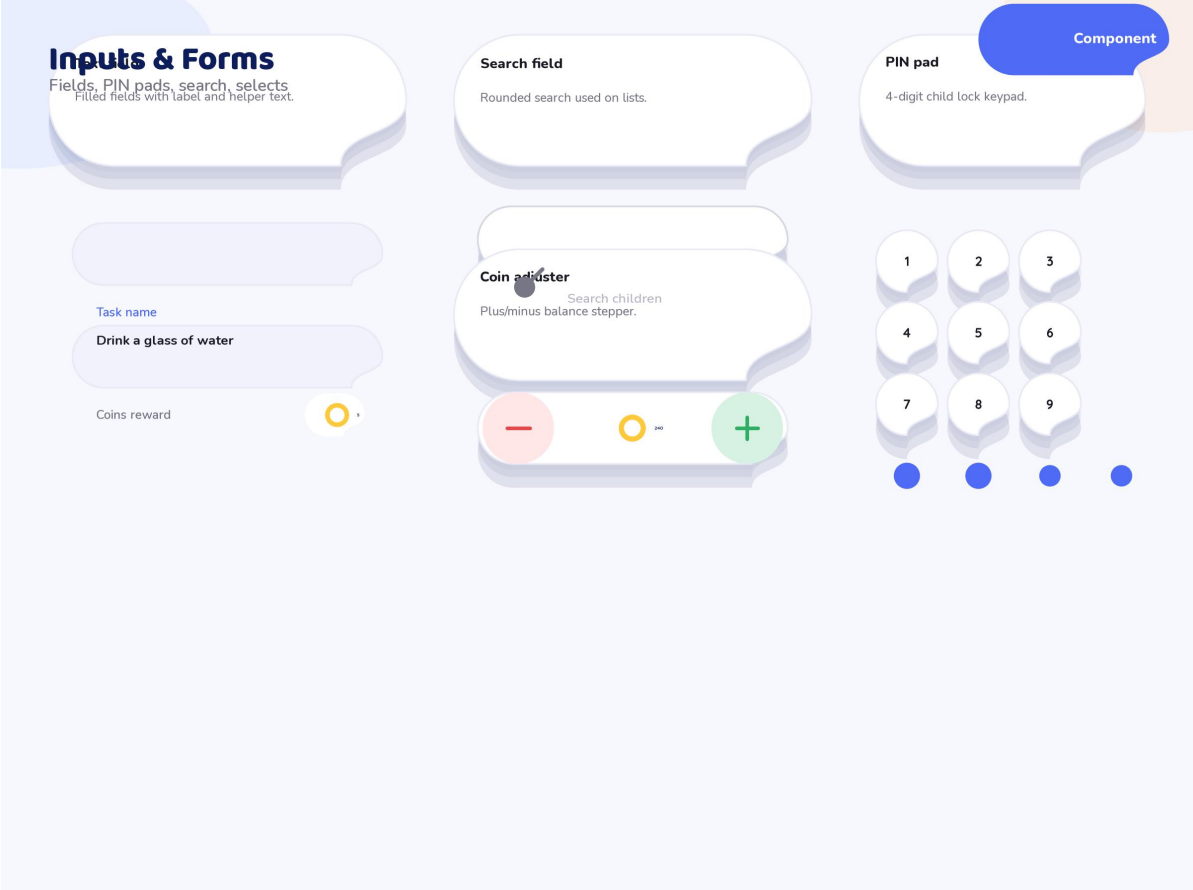
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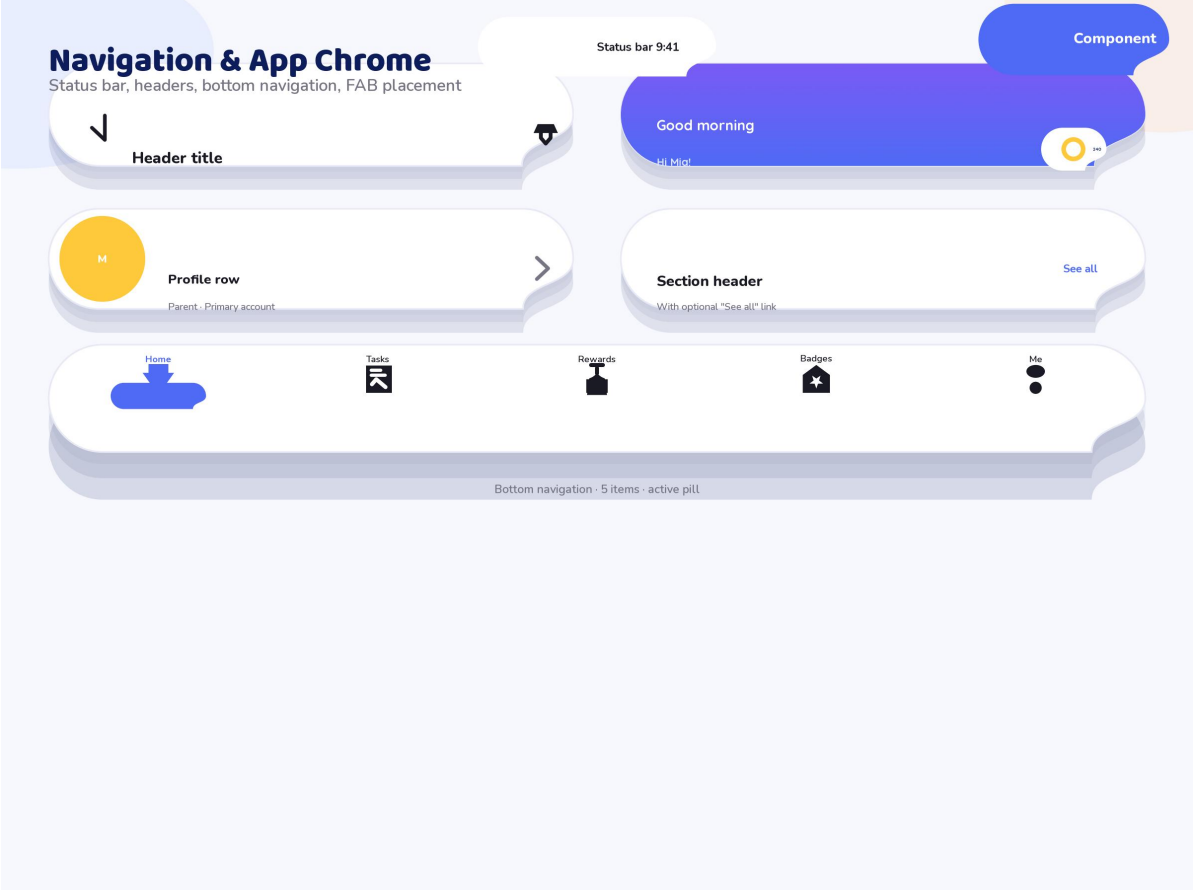
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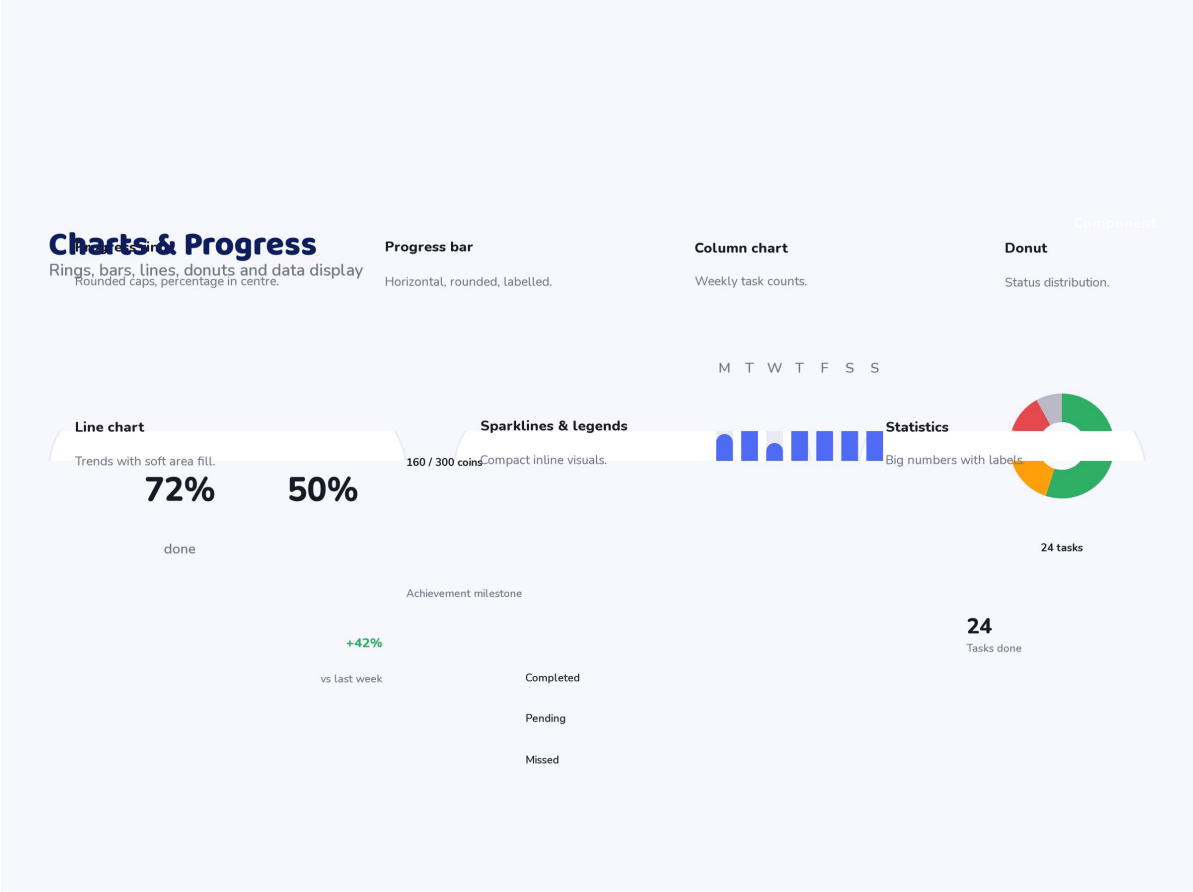
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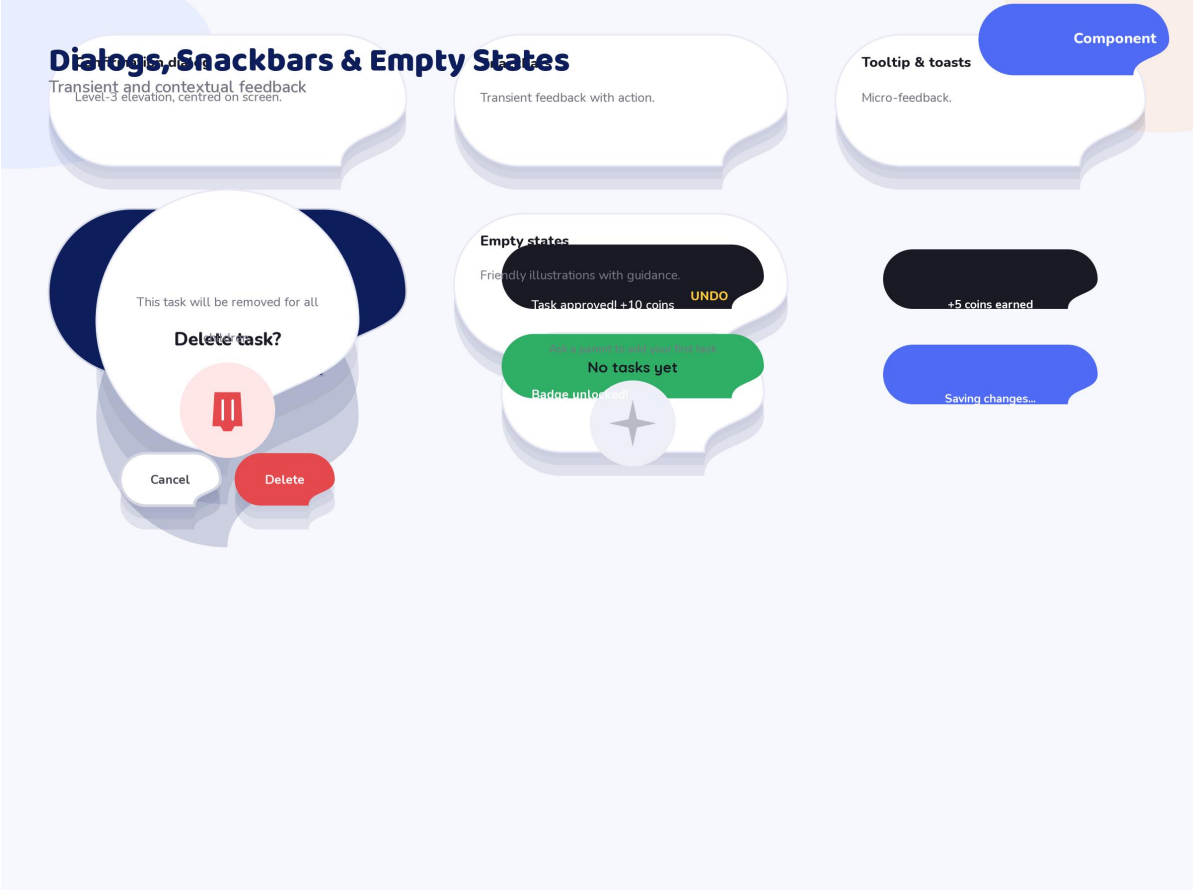
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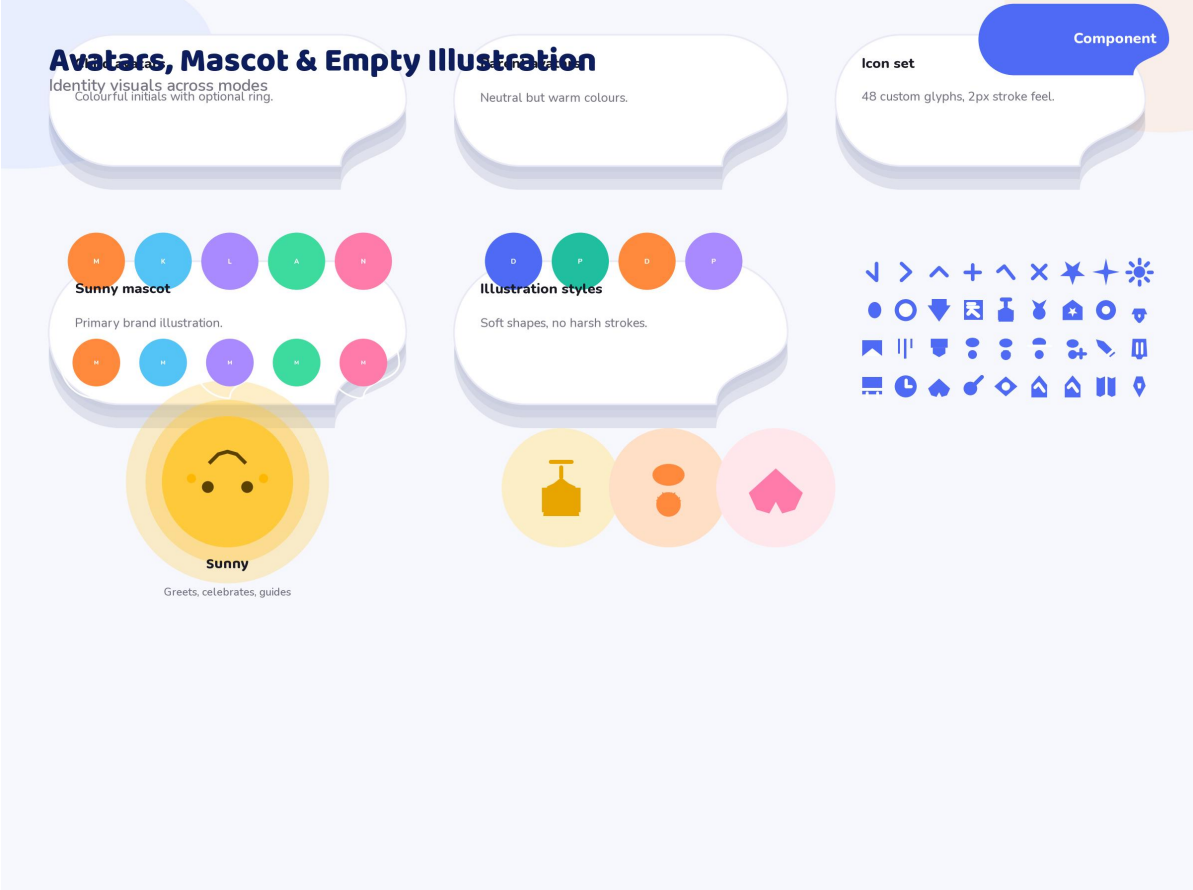
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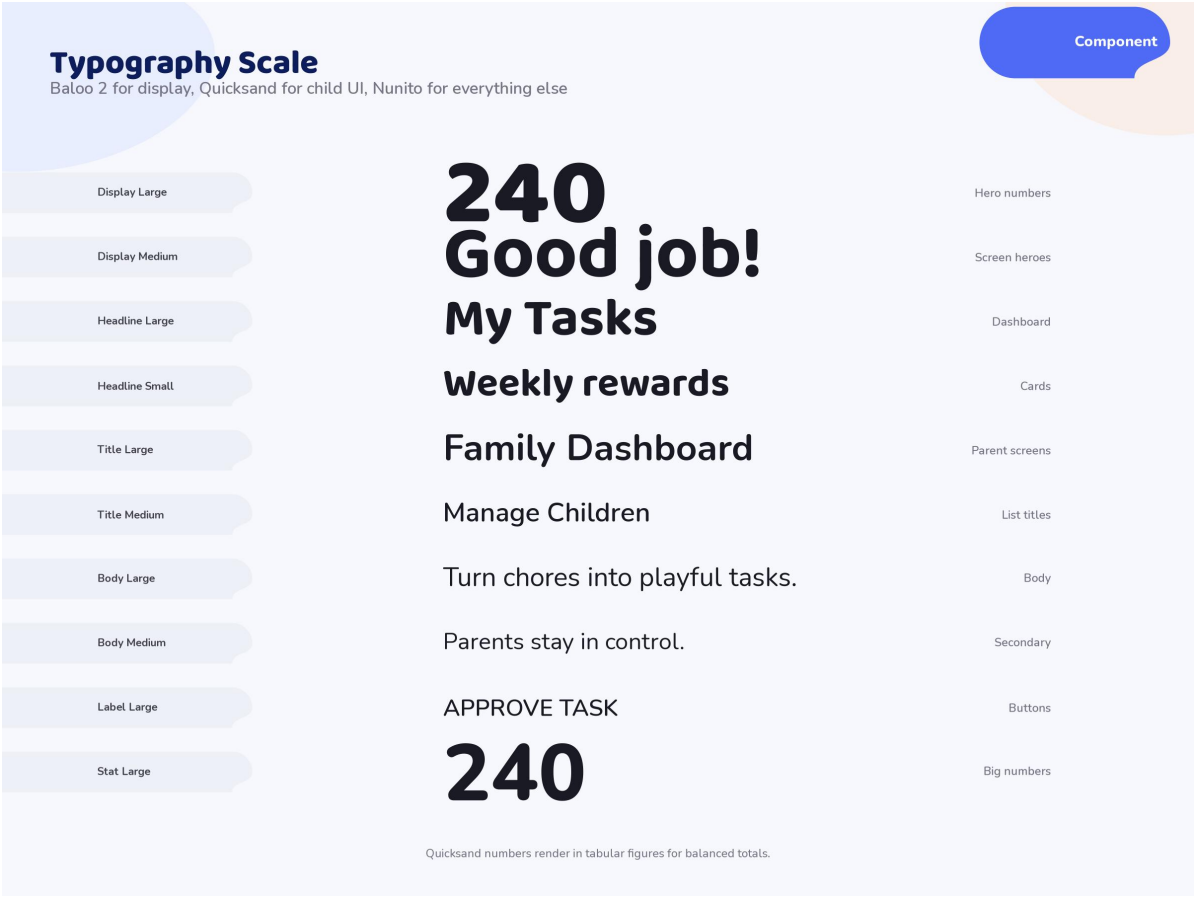
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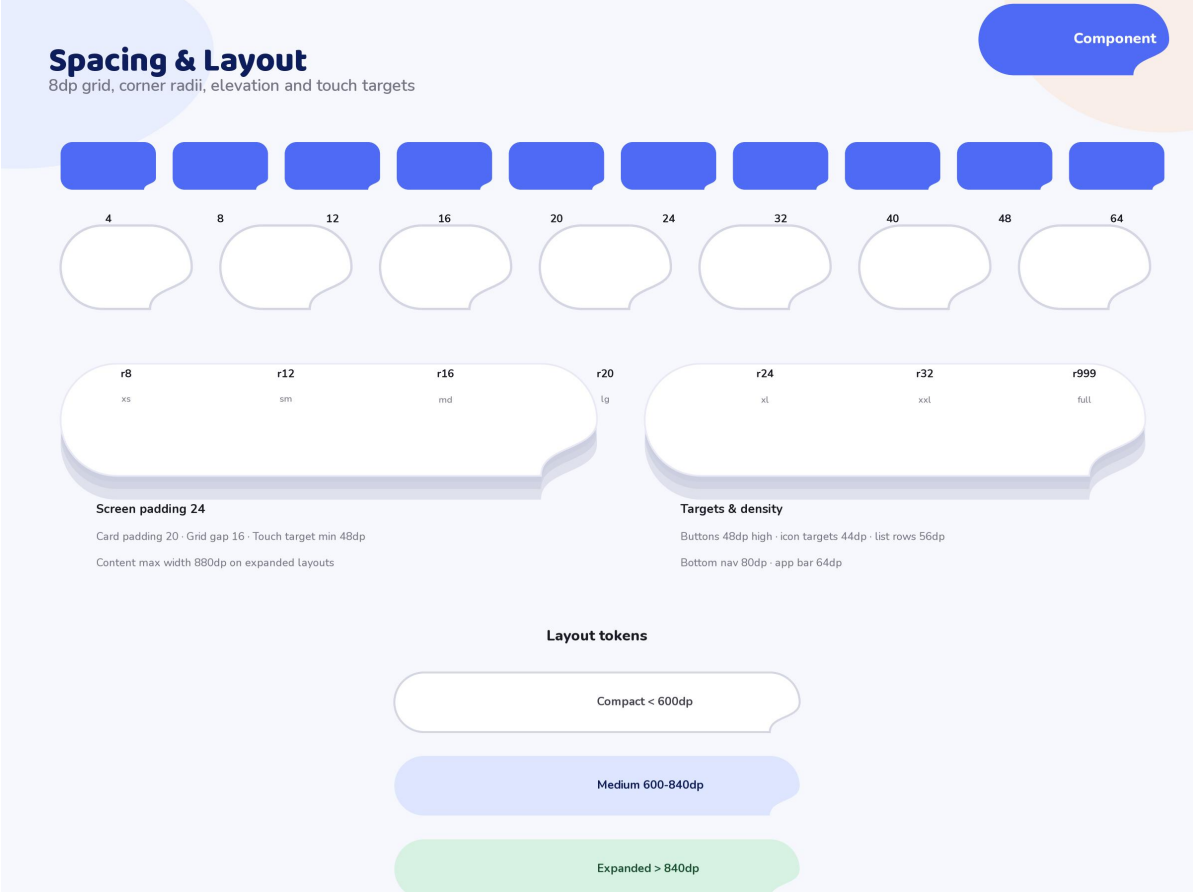
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11*: Colors – The full light and dark palettes with brand, accent, semantic and status colours and their exact hex values.*



12*: Spacing – The 4–64 px spacing scale, radii xs–full, elevation levels 0–4 and the 6/12 column grid with 16 px gutters.*

CHAPTER SEVEN

ICONS AND ILLUSTRATIONS

7.1 The Icon Set

Fifty-two custom SVG icons cover navigation, tasks, rewards, coins, badges, reports, messages and settings. Icons follow a 24 × 24 viewBox, 2 px stroke weight, rounded caps and joins, and use the onSurfaceVariant colour at rest. Each icon is exported as a single standalone SVG file under assets/icons/ with the sch_icon_ prefix.

13*: Icon categories and counts*

Category	Icons (naming root)	Count
Navigation	home, grid, list, back, close, search, menu	7
Tasks	check, clock, flag, lock, sparkle, star, calendar, repeat	8
Rewards & coins	coin, coins, gift, trophy, tag, cart, redeem	7
Badges	shield, award, medal, crown, diamond, flame, streak	7
Reports	chart, chart_line, chart_bar, chart_donut, download, export	6
People & profiles	person, person_add, child, parents, avatar, pin	6
Messages	chat, chat_dot, bell, mail, send	5
Settings & system	settings, tune, lock, theme, wifi, info, help, warning	8

7.2 Illustrations

Five SVG illustrations carry the personality of the product. Each is drawn in flat vector style with the brand palette and exports as a standalone SVG under assets/illustrations/.

14*: Illustration assets*

Asset	File	Used on
Sunny mascot	sch_ill_sunny_mascot.svg	Splash, dashboard, about
Empty state	sch_ill_empty_tasks.svg	Empty task lists
Celebration	sch_ill_celebration.svg	Badge unlocks, achievements
Welcome hero	sch_ill_welcome_hero.svg	Welcome screen
Error state	sch_ill_error_state.svg	Offline/error screens

7.3 Asset Conventions

- All assets carry the sch_ asset prefix so the whole family is discoverable.
- Icons ship as stroke SVGs; illustrations as flat-colour SVGs.
- SVGs are optimised, well-formed XML and render without external dependencies.
- Never rasterise SVG assets at design time; Flutter renders them natively.

CHAPTER EIGHT

MOTION AND ANIMATION

8.1 Principles

1. Purposeful – every animation explains a state change; nothing moves for decoration.
2. Fast – durations run 50–900 ms so the interface never feels slow.
3. Joyful – celebrations (coins, confetti, badge glow) reward success; errors animate calmly.

8.2 Durations and Curves

15: *Reference durations and easing curves*

Token	Duration	Curve
instant	50 ms	None (hard cut)
fast	150 ms	standard
normal	250 ms	standard
gentle	350 ms	emphasized
slow	500 ms	emphasized
extra	750 ms	bounce

16: *Easing curves*

Curve	Specification
standard	cubic-bezier(0.2, 0.0, 0.0, 1.0)
emphasized	cubic-bezier(0.3, 0.0, 0.1, 1.0)
decelerate	cubic-bezier(0.0, 0.0, 0.2, 1.0)
accelerate	cubic-bezier(0.4, 0.0, 1.0, 1.0)
spring	spring(weight: 0.9, damping: 14)
bounce	cubic-bezier(0.34, 1.56, 0.64, 1.0)

8.3 Signature Moments

17: *Signature animations specified for implementation*

Animation	Duration	Behaviour
Coin fly	450 ms	Coin pops at the task card, arcs to the balance chip, total counts up with bounce
Check draw	500 ms	Check mark draws itself; card pulses green; a coin flies
Confetti	750 ms	Bursts from the task card; particles fall and rotate

Badge unlock	600 ms	Badge scales 0 to 1 with a golden glow ring
Reward claim	500 ms	Reward card lifts, confetti at top, check stamp appears
Screen transition	300 ms	Fade-through with an 8% horizontal slide

8.4 Loading Patterns

Loading states use three bouncing dots or a gently pulsing spinning coin for 1000 ms loops. Never show a frozen frame; every asynchronous action has an animated placeholder.

CHAPTER NINE

RESPONSIVE BEHAVIOUR AND DARK MODE

9.1 What Changes at Each Tier

18*: Responsive playbook across breakpoints*

Element	Compact (< 600)	Medium (600–840)	Expanded (> 840)
Navigation	Bottom bar	Bottom bar	Rail + app bar
Grid columns	4	6	12
Screen padding	16 px	24 px	24 px + 880 max width
Dashboard cards	1 column	2 columns	3 columns
Charts	Compact labels	Full labels	Full labels + legend

9.2 Phone versus Tablet

Phone is usable but the product is optimised for tablet portrait (medium tier), the typical family-kiosk orientation. On compact screens, multi-column card grids collapse to single columns, charts reduce label density, and dialogs become full-width bottom sheets.

9.3 Dark Mode

The app respects a family-level theme setting (light, dark or follow the shared tablet). Both themes pass the same contrast minimums. Illustrations ship in both palettes or use theme-aware tokens so no asset looks wrong on the dark surfaces.

9.4 Rules to Remember

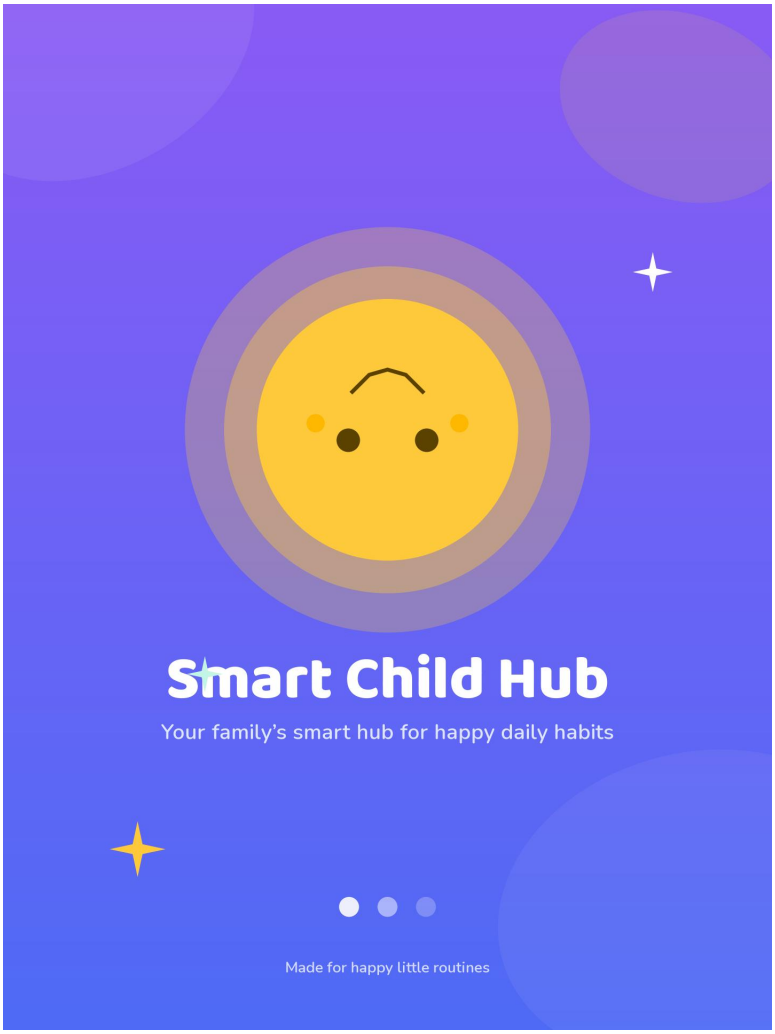
- Keep it simple at every size; the same content exists on all tiers, only the rhythm changes.
- Never hide an action behind an extra tap on smaller screens; collapse, do not remove.
- Respect the 48 px minimum touch target in every tier.

CHAPTER TEN

SCREEN MOCKUPS

Twenty-two full-screen mockups define the complete application at 1440 × 1920. Each screen below includes its mode, purpose and the key elements that must be implemented. The mockups are the authoritative visual contract for Phase 3 development.

01_splash · Splash · Child mode Brand reveal: Sunny the mascot spins in, the Smart Child Hub logo scales to rest, then the app transitions into the welcome screen. Child mode tone immediately.



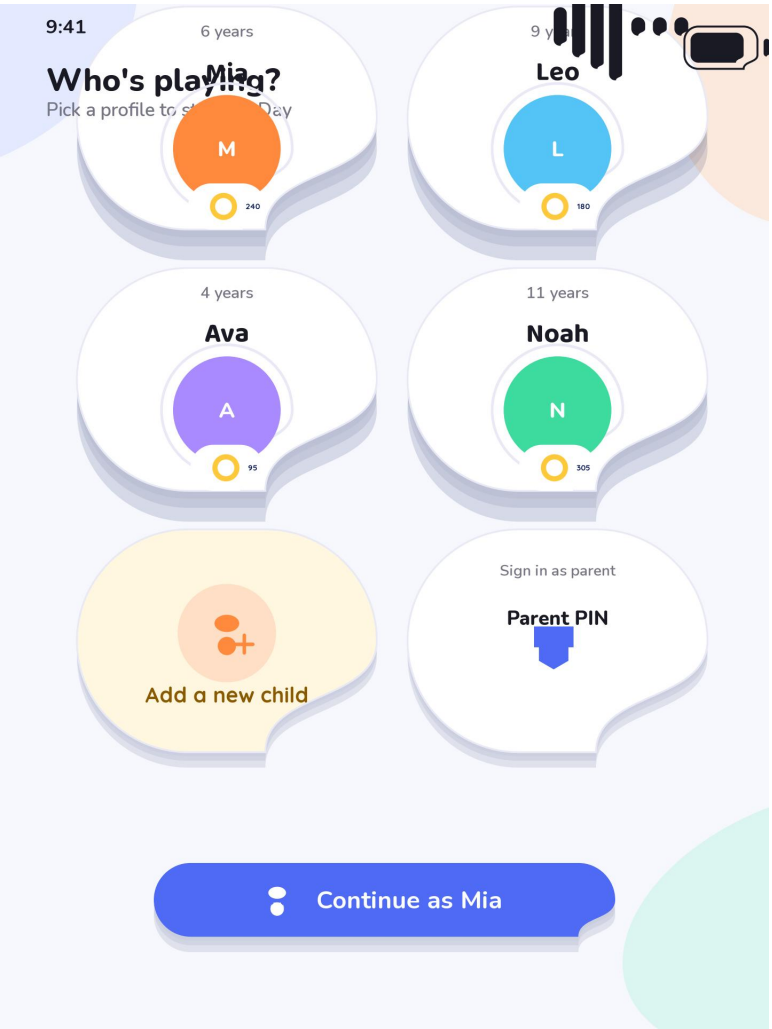
13*: Splash – Child mode*

02_welcome · Welcome · Shared mode Hero illustration, the product tagline and two clear entry points “I am a parent” and “I am a child”. No login is required for the child path.



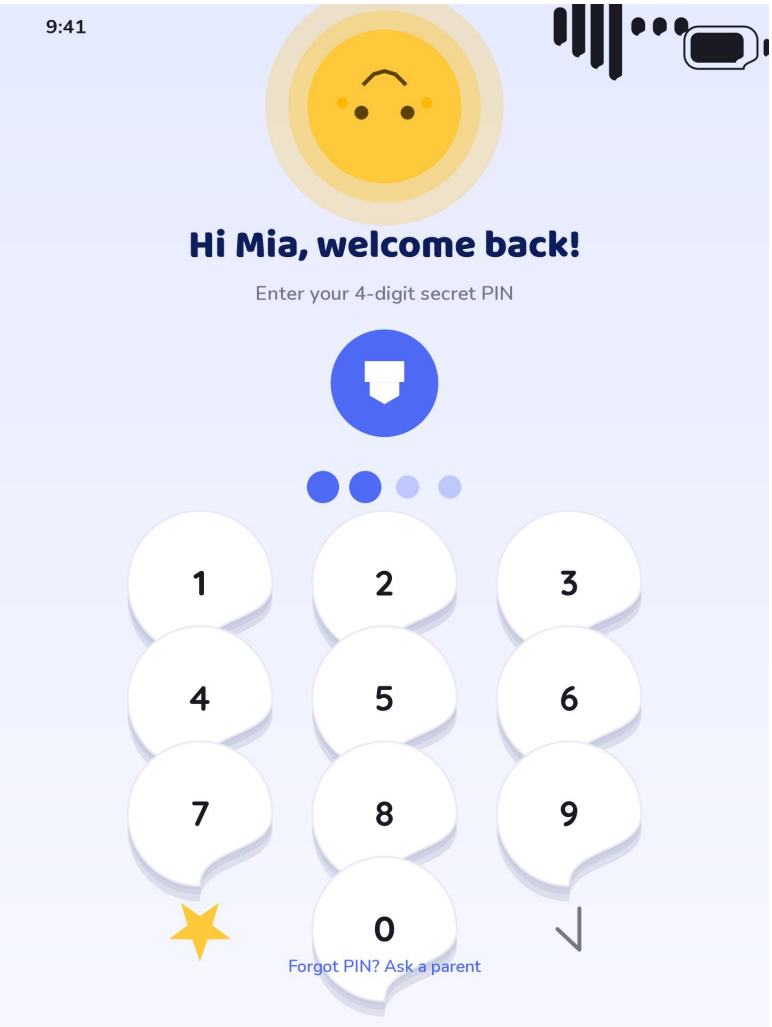
14: *Welcome – Shared mode*

03_choose_profile · Choose Profile · Child mode Kiosk profile picker: large round avatars with names, an "Add new profile" tile, and a parent-lock affordance. Profiles are touch-first and 96 px tall.



15: *Choose Profile – Child mode*

04_child_pin · Child PIN · Child mode Optional four-digit PIN gate protecting a specific child profile. Numeric keypad with large keys, gentle shake on error and a back affordance.



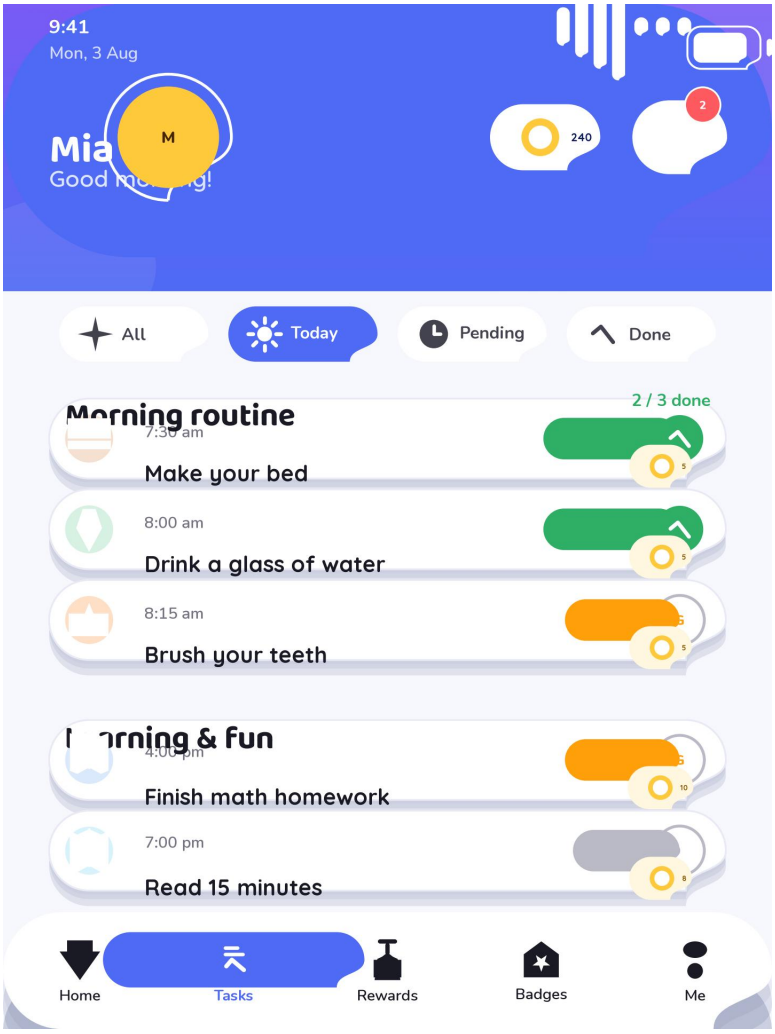
16*: Child PIN – Child mode*

05_child_dashboard · Child Dashboard · Child mode The daily hub in child mode: greeting with Sunny, today's progress ring, a task checklist with status pills, a coins balance chip and the bottom navigation bar.



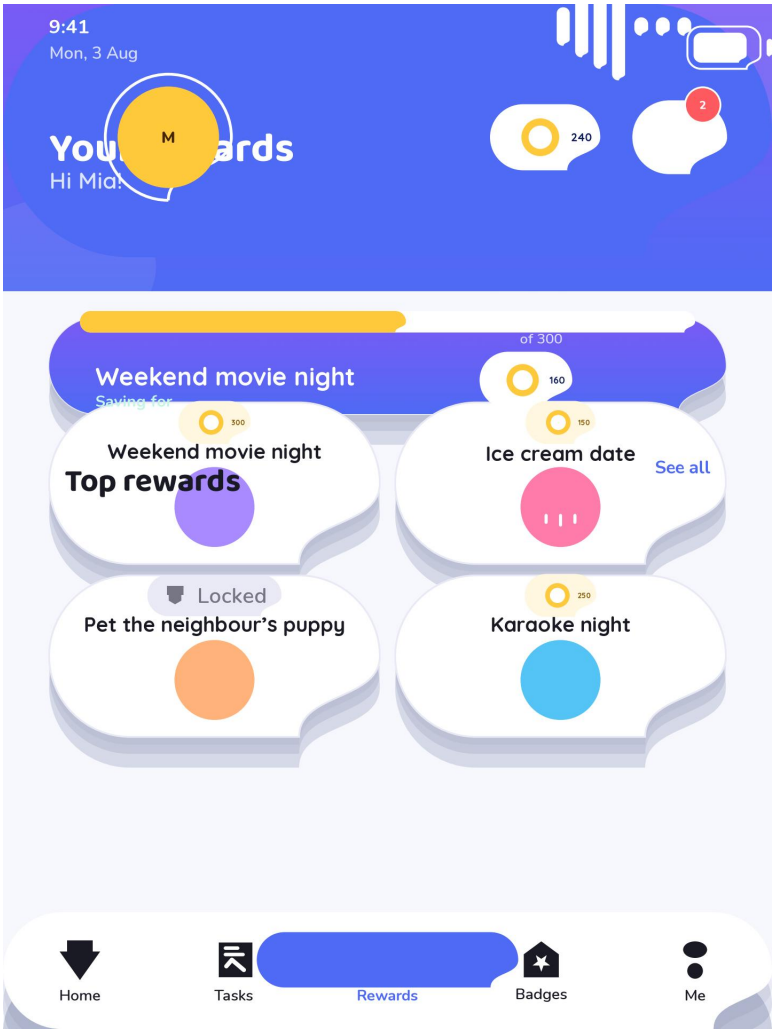
17: *Child Dashboard – Child mode*

06_task_screen · Task Screen · Child mode Task detail with a bright illustration, step hints, a big completed button and a coin reward badge. Supports self-check and parent-approval variants.



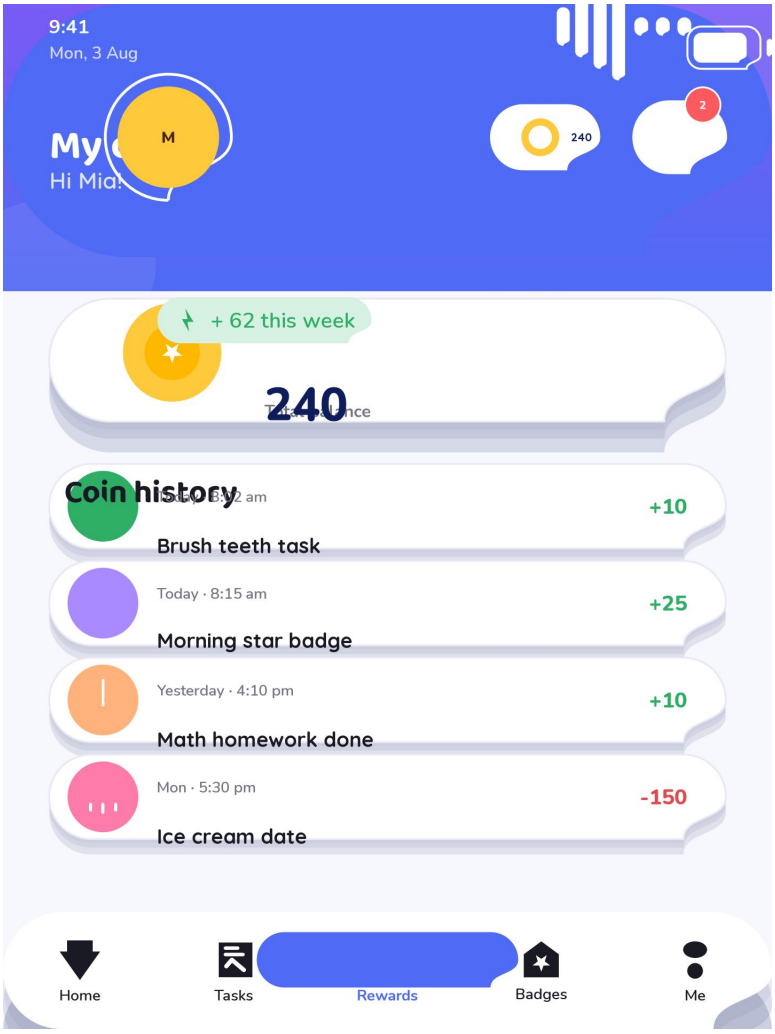
18: *Task Screen – Child mode*

07_rewards · Rewards · Child mode Rewards shop for the child: reward cards with illustrations, coin prices, available/claimed states and a confirmation dialog before a redemption is submitted.



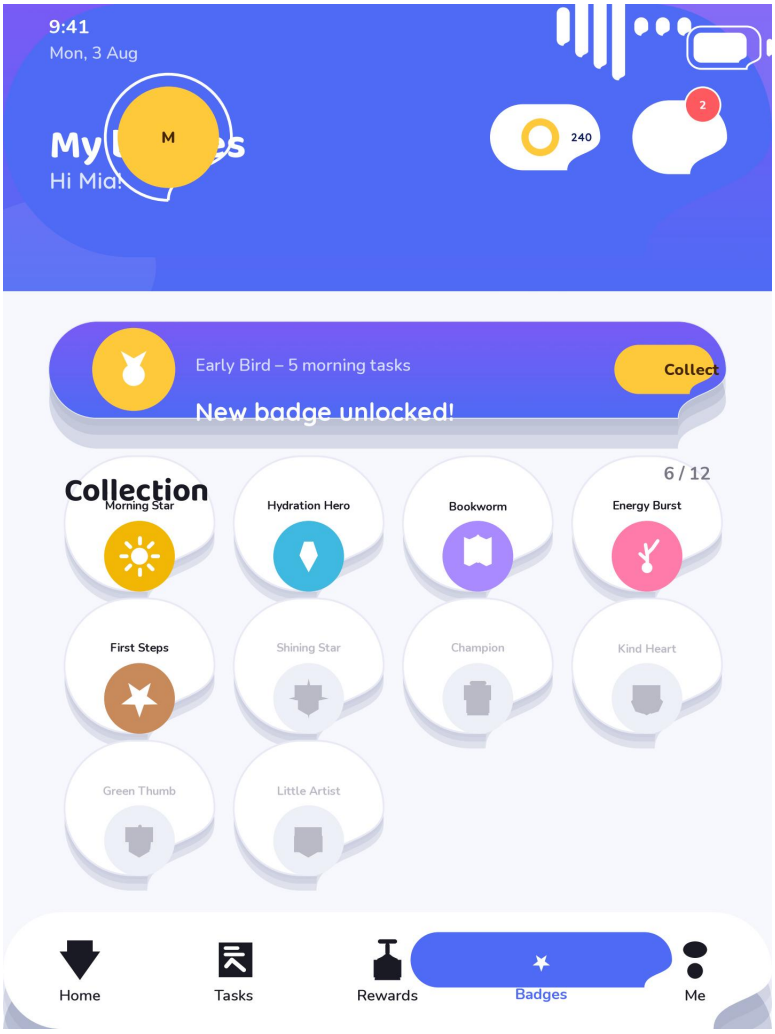
19: *Rewards – Child mode*

08_coins · Coins · Child mode Coin ledger: current balance card, earn/spend breakdown, a daily history list with icons and the running total in tabular figures.



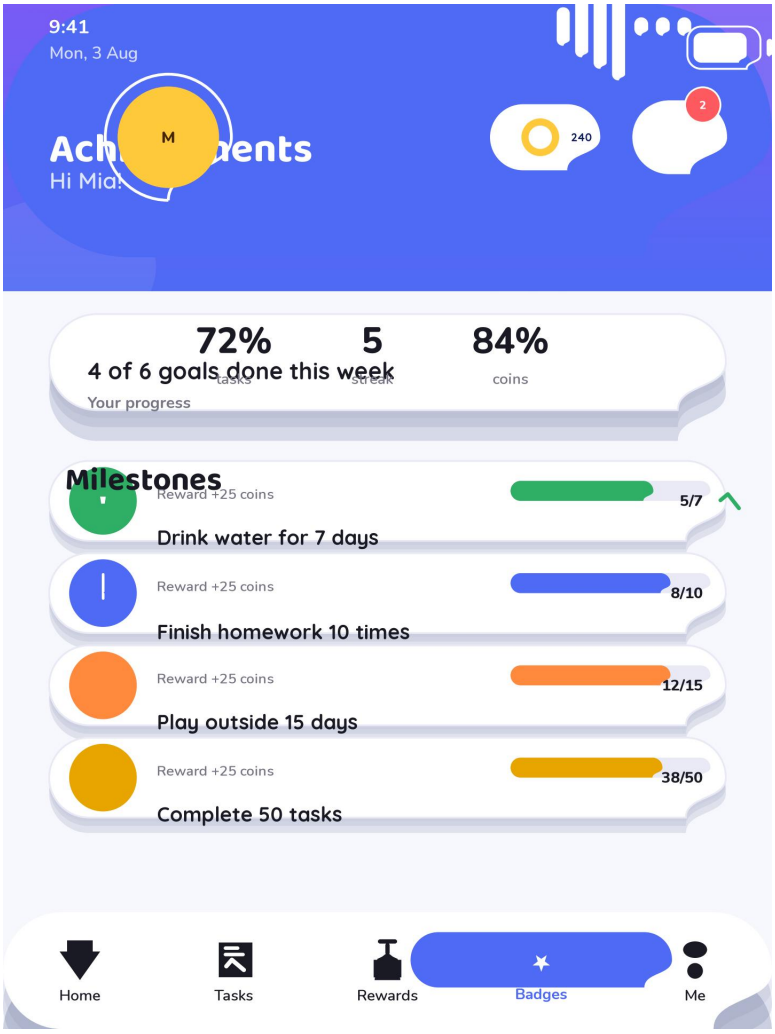
20*: Coins – Child mode*

09_badges · Badges · Child mode Badge collection in child mode: bronze, silver, gold and diamond tiers with locked and unlocked states and progress hints.



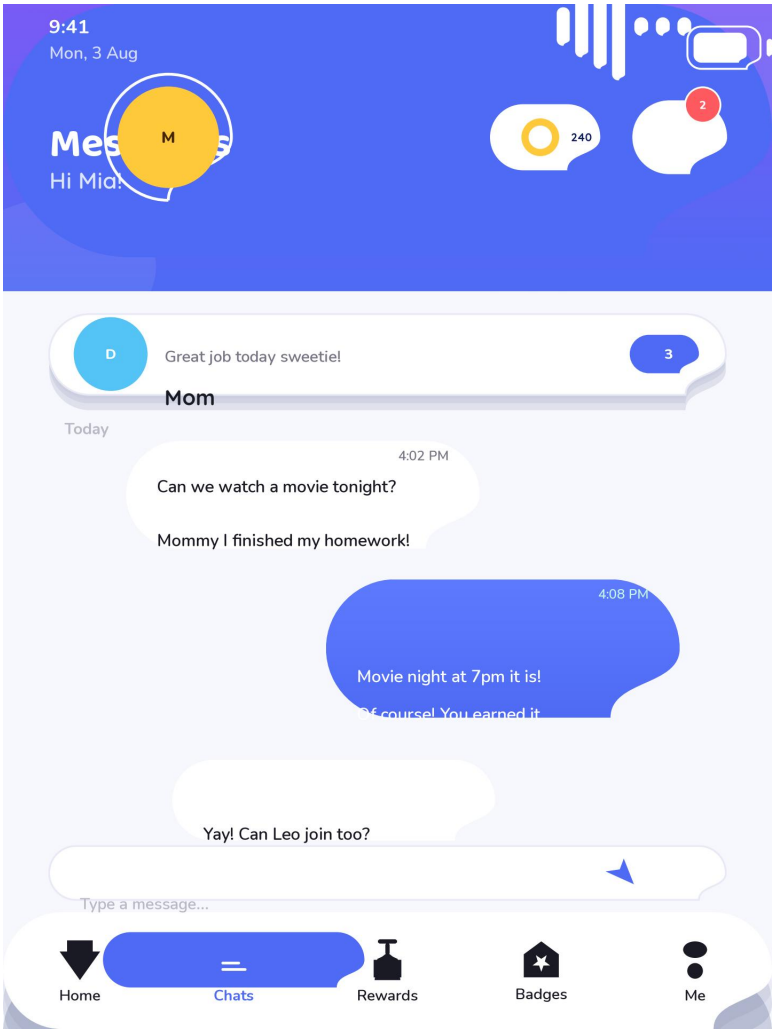
21: *Badges – Child mode*

10_achievements · Achievements · Child mode Streak and milestone view: week strip with completion days, milestone progress bars, confetti moment when a new tier is reached.



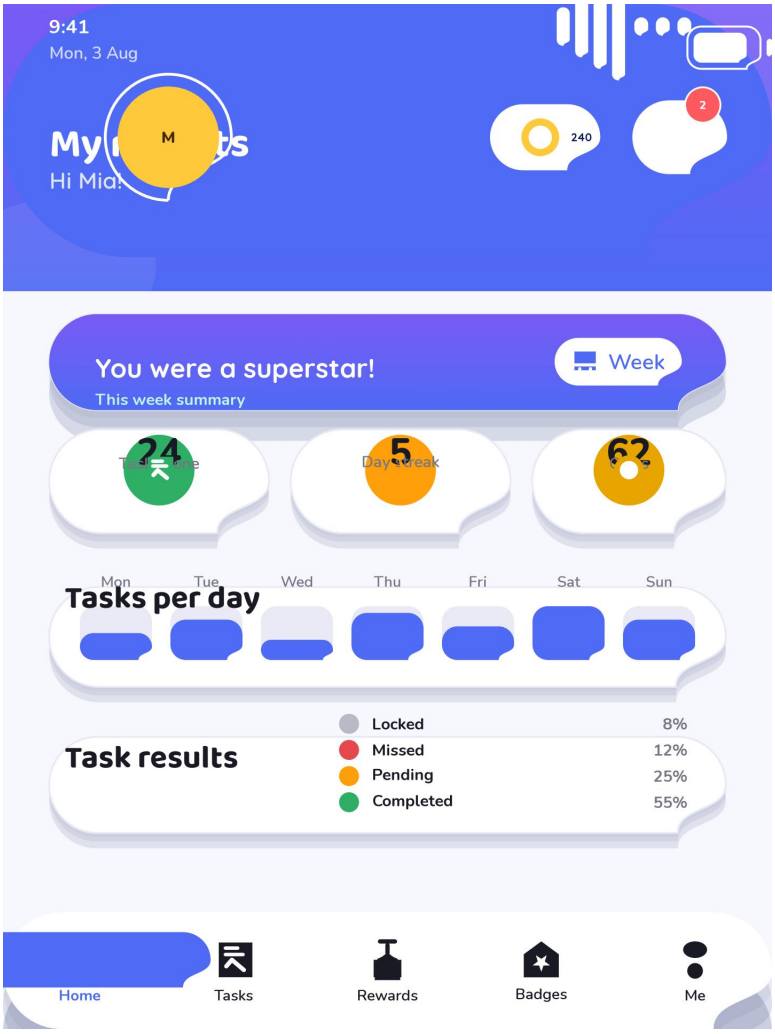
22: *Achievements – Child mode*

11_messages · Messages · Child mode Parent-child messages: simple positive notes with big icons, read states and a quick-reply composer. Encourages communication without a phone.



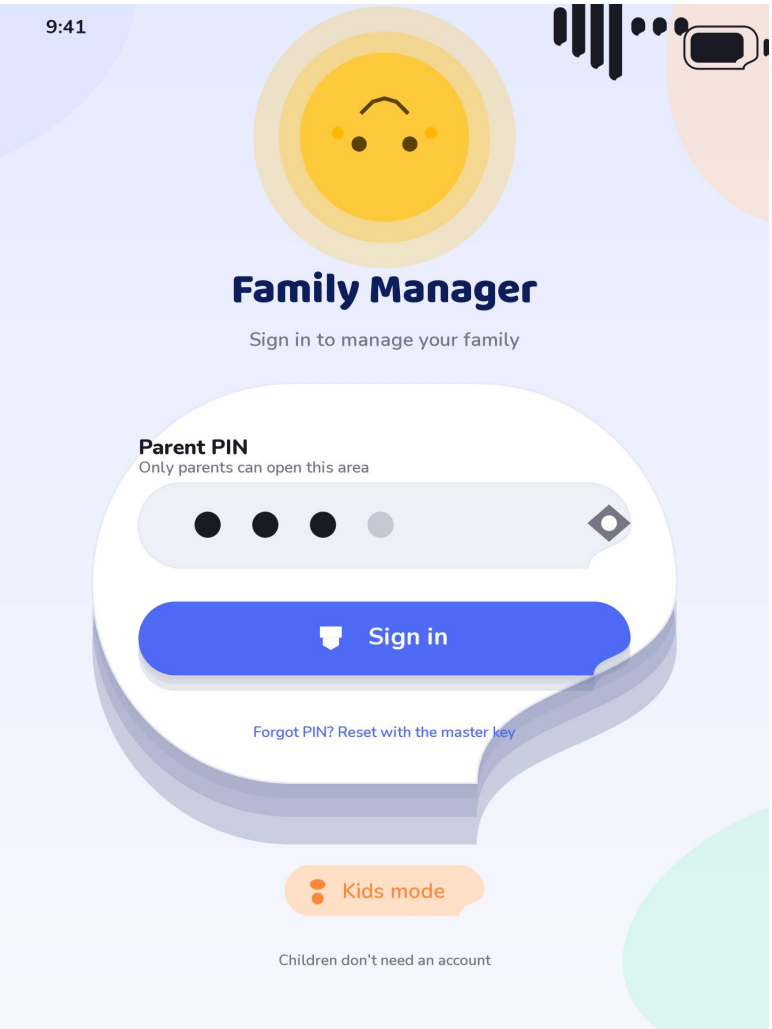
23: Messages – Child mode

12_reports_child · Child Reports · Child mode Child-facing summary: weekly completion percentage, coins earned, badges gained and a simple line chart of the last seven days.



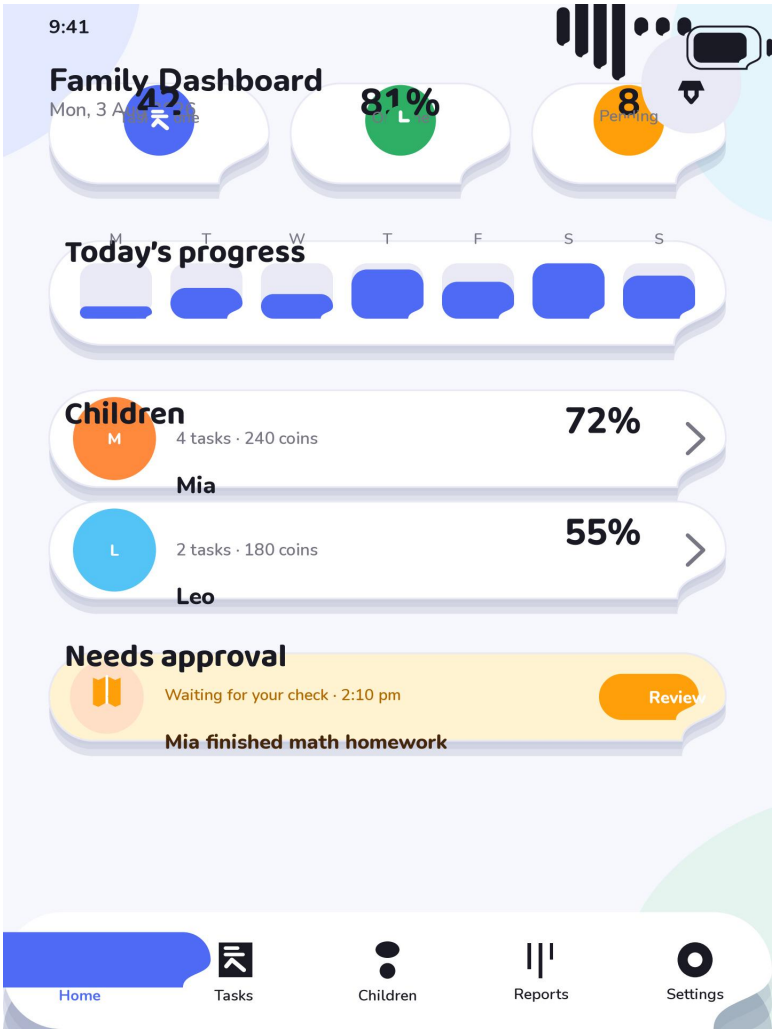
24: Child Reports – Child mode

13_parent_login · Parent Login · Parent mode Parent access gate: PIN pad in the professional parent theme, wrong-PIN lockout messaging and a hint about recovery.



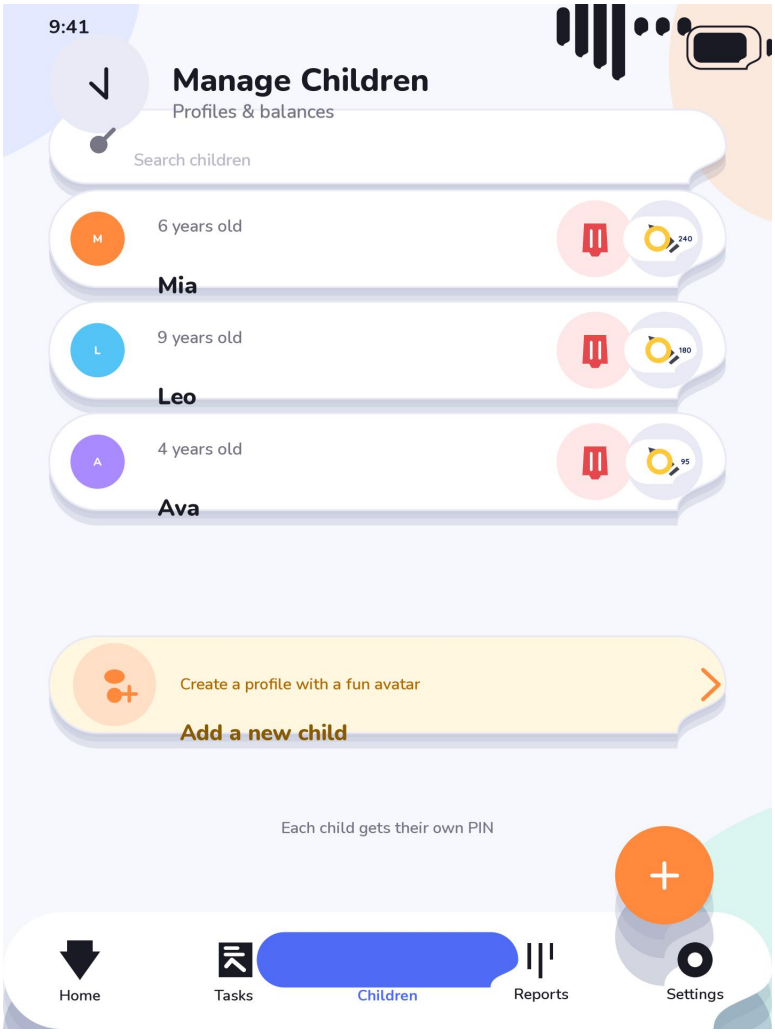
25: *Parent Login – Parent mode*

14_parent_dashboard · Parent Dashboard · Parent mode Family manager overview: child summary cards with completion rates, pending approvals badge, coin circulation totals and a navigation rail.



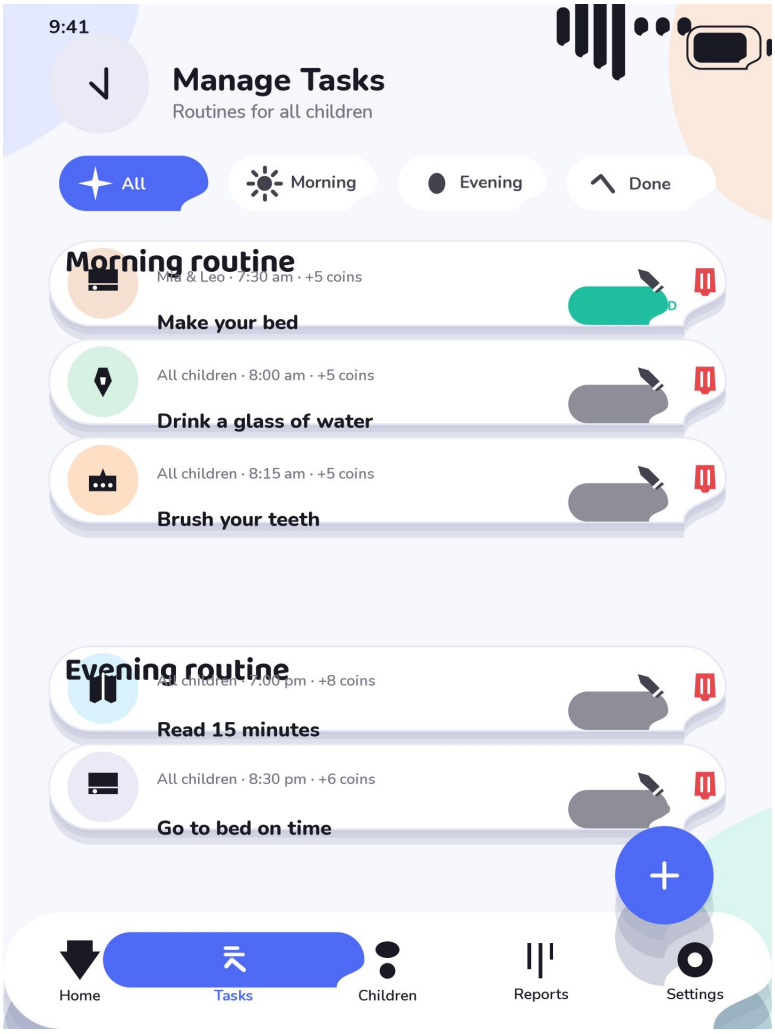
26: Parent Dashboard – Parent mode

15_manage_children · Manage Children · Parent mode Parent administration of profiles: add/edit/remove children, avatars, PINs and active/inactive state. Full-row touch targets in the parent theme.



27: Manage Children – Parent mode

16_manage_tasks · Manage Tasks · Parent mode Task management: list of tasks with frequency, coin value, verification type and assignee chips; add, edit, duplicate and archive actions.



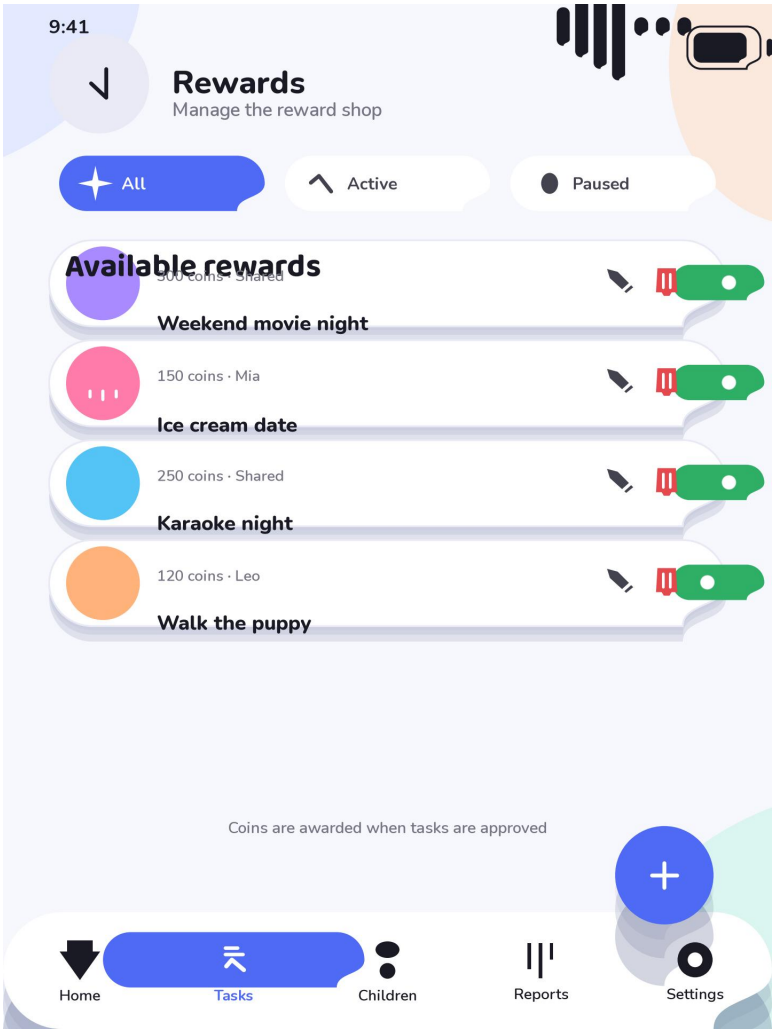
28: Manage Tasks – Parent mode

17_pending_approval · Pending Approval · Parent mode Approval queue: each card shows the child's submission with proof photo thumbnail, approve and reject buttons and reason input on reject.



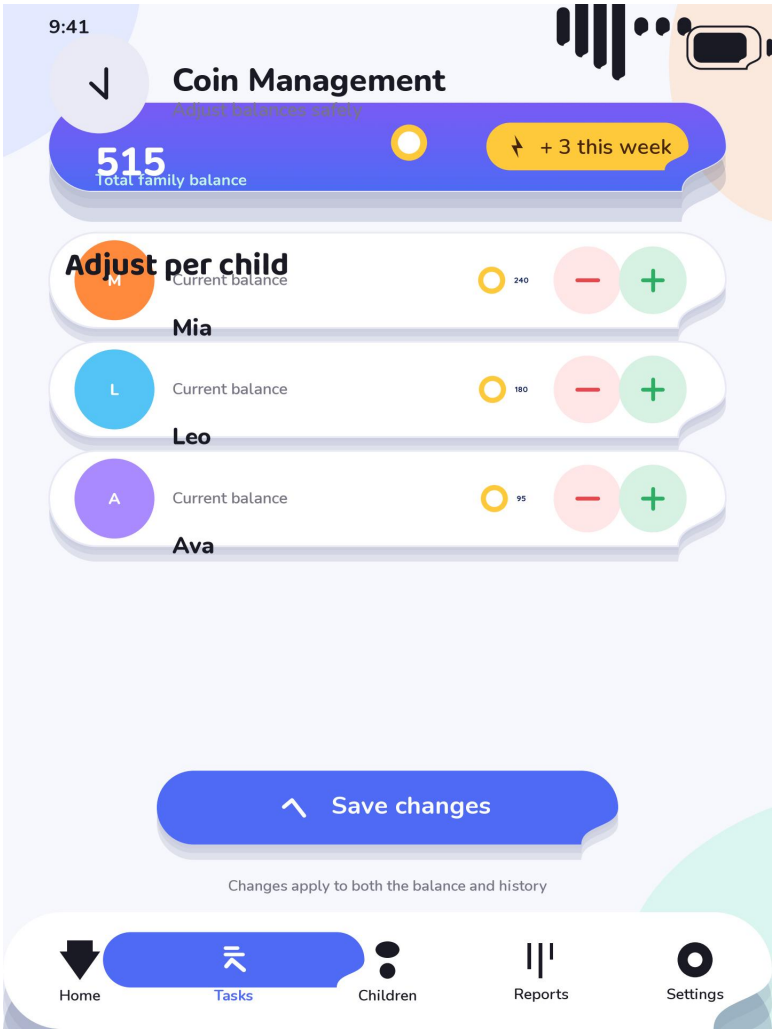
29: Pending Approval – Parent mode

18_rewards_management · Rewards Management · Parent mode Rewards administration: create rewards with illustration, coin price and stock; review redemption requests and approve or decline them.



30: *Rewards Management – Parent mode*

19_coin_management · Coin Management · Parent mode Coin ledger for the parent: per-child balances, manual adjust, transaction history filters and export. Tabular numerals keep totals stable.



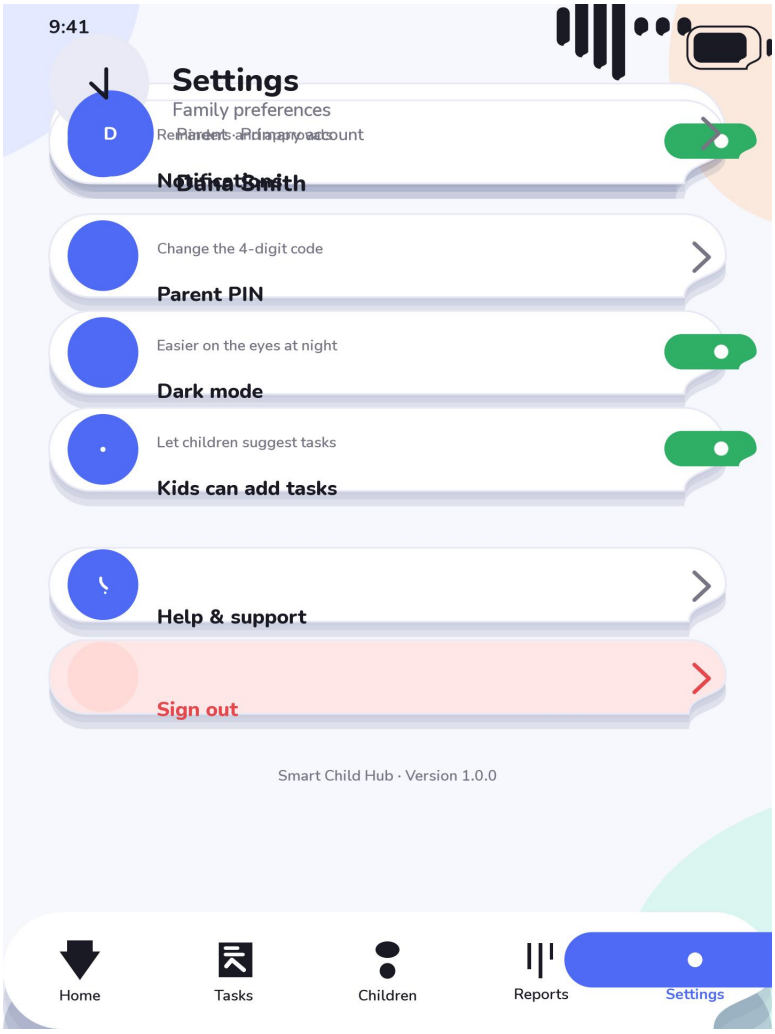
31: *Coin Management – Parent mode*

20_reports_parent · Parent Reports · Parent mode Period reports: daily, weekly, monthly and yearly tabs with completion rates, missed tasks, top-child ranking and export to PDF.



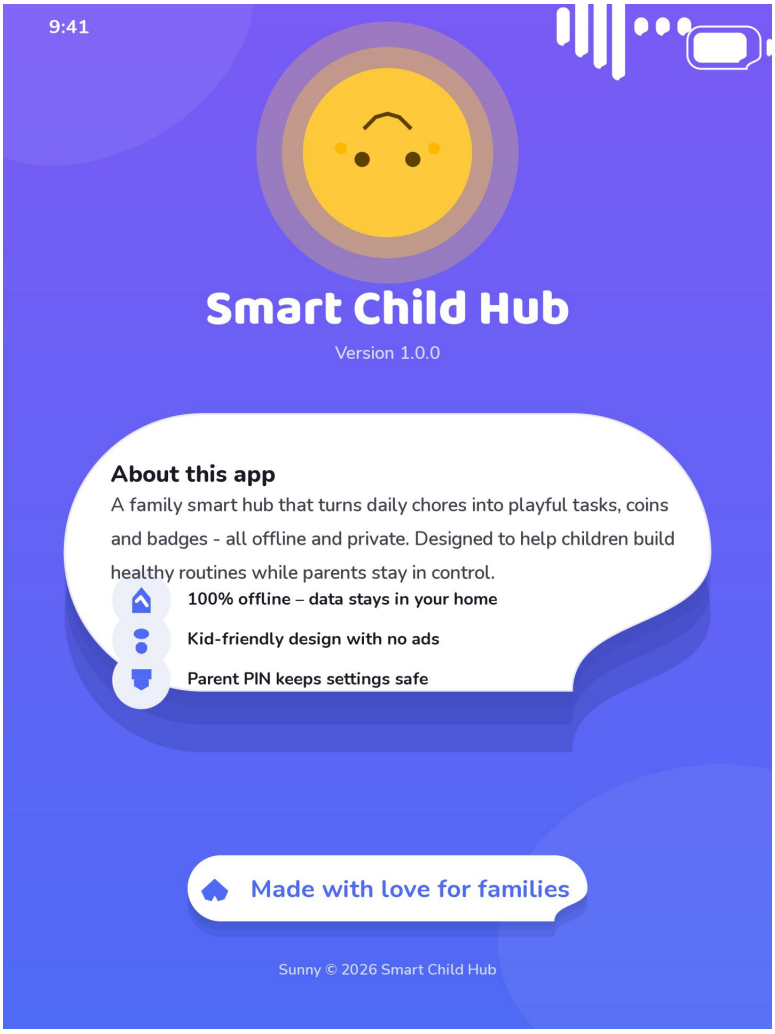
32: *Parent Reports – Parent mode*

21_settings · Settings · Parent mode Family settings: schedule defaults, verification rules, notification preferences, theme (light/dark) and data backup within the home network.



33*: Settings – Parent mode*

22_about · About · Parent mode About screen: product story, version, credits, the offline/privacy statement and a link to this design system.



34: *About – Parent mode*

CHAPTER ELEVEN

NAMING CONVENTIONS AND ASSET STRUCTURE

11.1 Naming Rules

19: *Naming conventions*

Element	Case	Example
Asset files	snake_case with sch_ prefix	sch_icon_coin.svg
Classes and widgets	UpperCamelCase	RewardsScreen
Constants and tokens	UPPER_SNAKE_CASE	APP_PADDING = 24
Screen files	snake_case with _screen suffix	rewards_screen.dart
Widget files	snake_case with _widget suffix	coin_chip_widget.dart
Colour tokens	snake_case with _color suffix	success_color

11.2 Folder Structure

The design deliverables are organised so that every artefact has one obvious home. The Phase 3 Flutter project should mirror this logic in lib/theme, lib/widgets, lib/screens and lib/assets.

- documentation/design-system – this guide and the screen specifications
- documentation/style-guide – the seven PDF guides
- documentation/mockups – the twenty-two screen mockups
- documentation/components – the eleven component boards
- documentation/assets/icons – the fifty-two SVG icons
- documentation/assets/illustrations – the five SVG illustrations
- documentation/scripts – the token source and all generator scripts

11.3 Token to Flutter Theme Mapping

Design tokens map to a Material 3 ThemeData so implementation needs no design decisions. The primary seed is Color(0xFF4F6BF5); TextTheme maps display styles to Baloo 2 and body styles to Nunito; fontFamily for numerals uses Quicksand with tabular figures. The complete mapping is documented in the Widget Guide PDF.

CHAPTER TWELVE

VALIDATION AND HANDOFF

12.1 Quality Checklist

1. Every screen exists as a mockup and every interactive element as a component board.
2. All tokens are defined once in scripts/tokens.py and used by every renderer.
3. Contrast minimums and 48 px touch targets hold across both themes.
4. All PDFs, SVGs and PNGs open cleanly and are checked for missing glyphs or corrupt pages.
5. Every naming convention in Chapter 11 is applied to the generated assets.

12.2 Handoff to Phase 3

With this design system complete, Phase 3 can begin Flutter development with zero ambiguity. Developers start from scripts/tokens.py to build the ThemeData, then implement the eleven component families against their boards, and finally compose the twenty-two screens from the mockups. The Widget Guide provides the direct component-to-widget translation and the recommended packages for fonts, animation and secure storage.

APPENDIX A

FILE INVENTORY

20: *Phase 2 file inventory*

Folder	File	Size
mockups	01_splash.png	102 KB
mockups	02_welcome.png	184 KB
mockups	03_choose_profile.png	246 KB
mockups	04_child_pin.png	178 KB
mockups	05_child_dashboard.png	280 KB
mockups	06_task_screen.png	259 KB
mockups	07_rewards.png	235 KB
mockups	08_coins.png	227 KB
mockups	09_badges.png	287 KB
mockups	10_achievements.png	252 KB
mockups	11_messages.png	180 KB
mockups	12_reports_child.png	257 KB
mockups	13_parent_login.png	164 KB
mockups	14_parent_dashboard.png	244 KB
mockups	15_manage_children.png	204 KB
mockups	16_manage_tasks.png	253 KB
mockups	17_pending_approval.png	241 KB
mockups	18_rewards_management.png	214 KB
mockups	19_coin_management.png	219 KB
mockups	20_reports_parent.png	232 KB
mockups	21_settings.png	208 KB
mockups	22_about.png	204 KB
components	01_buttons.png	334 KB
components	02_cards.png	215 KB
components	03_chips.png	247 KB
components	04_inputs.png	183 KB
components	05_navigation.png	180 KB
components	06_charts.png	148 KB
components	07_dialogs.png	250 KB
components	08_avatars.png	244 KB
components	09_typography.png	239 KB
components	10_colors.png	208 KB
components	11_spacing.png	179 KB
assets/icons	sch_icon_add_child.svg	0 KB
assets/icons	sch_icon_badge.svg	0 KB

assets/icons	sch_icon_bed.svg	0 KB
assets/icons	sch_icon_bell.svg	0 KB
assets/icons	sch_icon_book.svg	0 KB
assets/icons	sch_icon_calendar.svg	0 KB
assets/icons	sch_icon_chart.svg	0 KB
assets/icons	sch_icon_check.svg	0 KB
assets/icons	sch_icon_chev_down.svg	0 KB
assets/icons	sch_icon_chev_left.svg	0 KB
assets/icons	sch_icon_chev_right.svg	0 KB
assets/icons	sch_icon_child.svg	0 KB
assets/icons	sch_icon_clock.svg	0 KB
assets/icons	sch_icon_coin.svg	0 KB
assets/icons	sch_icon_cross.svg	0 KB
assets/icons	sch_icon_download.svg	0 KB
assets/icons	sch_icon_edit.svg	0 KB
assets/icons	sch_icon_eye.svg	0 KB
assets/icons	sch_icon_food.svg	0 KB
assets/icons	sch_icon_gear.svg	0 KB
assets/icons	sch_icon_heart.svg	0 KB
assets/icons	sch_icon_help.svg	0 KB
assets/icons	sch_icon_home.svg	0 KB
assets/icons	sch_icon_homework.svg	0 KB
assets/icons	sch_icon_info.svg	0 KB
assets/icons	sch_icon_lightning.svg	0 KB
assets/icons	sch_icon_lock.svg	0 KB
assets/icons	sch_icon_mail.svg	0 KB
assets/icons	sch_icon_medal.svg	0 KB
assets/icons	sch_icon_message.svg	0 KB
assets/icons	sch_icon_minus.svg	0 KB
assets/icons	sch_icon_moon.svg	0 KB
assets/icons	sch_icon_music.svg	0 KB
assets/icons	sch_icon_paint.svg	0 KB
assets/icons	sch_icon_parent.svg	0 KB
assets/icons	sch_icon_person.svg	0 KB
assets/icons	sch_icon_pets.svg	0 KB
assets/icons	sch_icon_plants.svg	0 KB
assets/icons	sch_icon_plus.svg	0 KB
assets/icons	sch_icon_run.svg	0 KB
assets/icons	sch_icon_search.svg	0 KB
assets/icons	sch_icon_send.svg	0 KB

assets/icons	sch_icon_shield.svg	0 KB
assets/icons	sch_icon_sparkle.svg	0 KB
assets/icons	sch_icon_star.svg	0 KB
assets/icons	sch_icon_sun.svg	0 KB
assets/icons	sch_icon_target.svg	0 KB
assets/icons	sch_icon_tasks.svg	0 KB
assets/icons	sch_icon_trash.svg	0 KB
assets/icons	sch_icon_trophy.svg	0 KB
assets/icons	sch_icon_verified.svg	0 KB
assets/icons	sch_icon_water.svg	0 KB
assets/illustrations	sch_ill_celebration.svg	0 KB
assets/illustrations	sch_ill_empty_tasks.svg	0 KB
assets/illustrations	sch_ill_error_state.svg	0 KB
assets/illustrations	sch_ill_sunny_mascot.svg	0 KB
assets/illustrations	sch_ill_welcome_hero.svg	1 KB
style-guide	Animation_Guide.pdf	149 KB
style-guide	Color_Palette.pdf	144 KB
style-guide	Component_Library.pdf	2513 KB
style-guide	Responsive_Guide.pdf	136 KB
style-guide	Typography_Guide.pdf	176 KB
style-guide	UI_Style_Guide.pdf	410 KB
style-guide	Widget_Guide.pdf	182 KB
design-system	Design_System_Guide.docx	this document

End of Design System Guide. Version 1.0.0 · August 2026